

Point Patterns II

HES 505 Fall 2025: Session 16

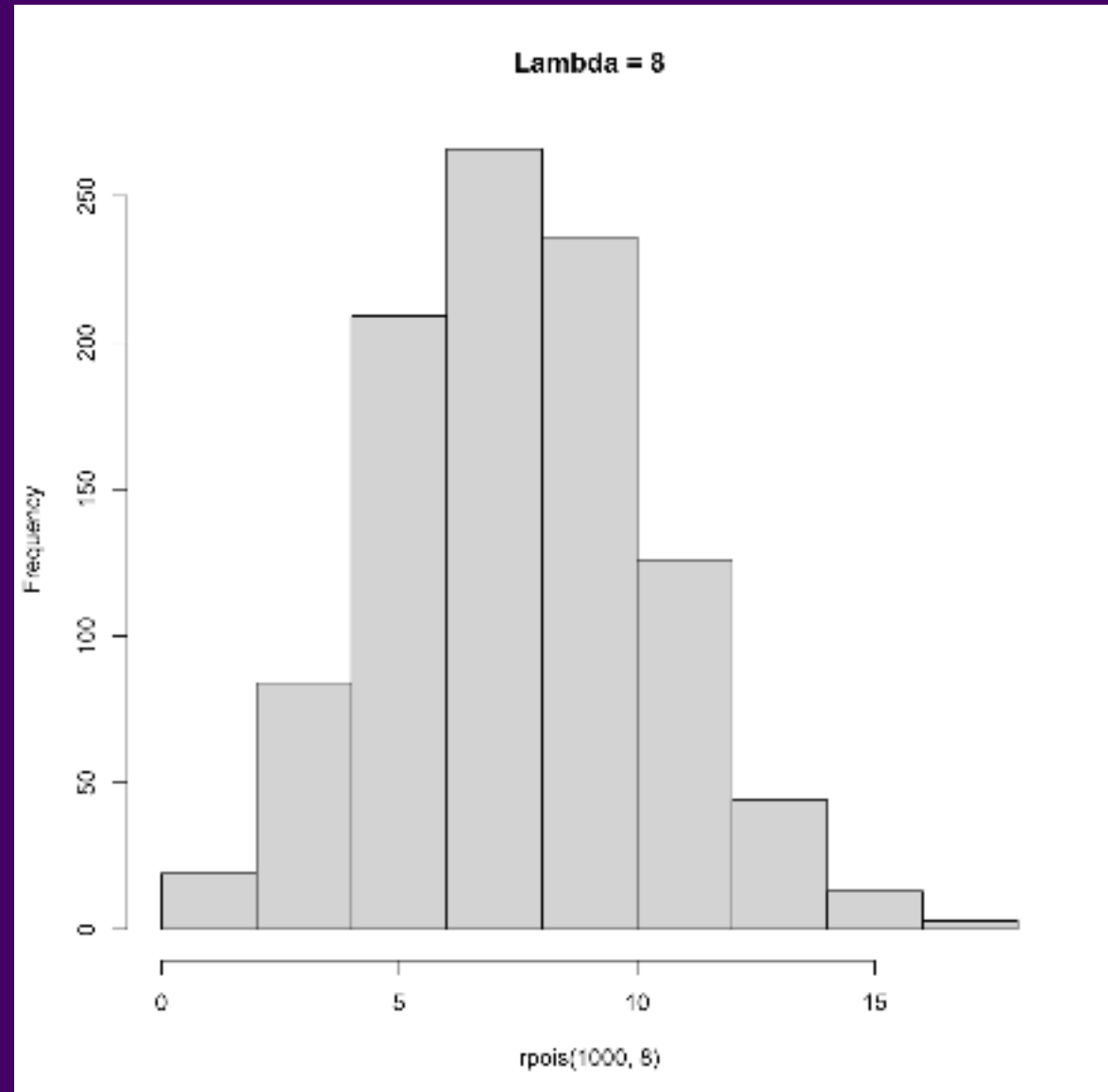
Matt Williamson

Point Processes and Why You Should Care

- They're ubiquitous
- Well-specified to many social and ecological theories
- Helps build intuition for interpolation

The Poisson Distribution

- Discrete probability distribution
- Probability of events in a fixed interval of time or space
- Mean (expectation) = variance = λ



Re-visiting CSR

What is CSR? What is it for?

What is the business case for CSR?

What is the moral case for CSR?

What is the role of the state?

What is the role of civil society?

What is the role of the media?

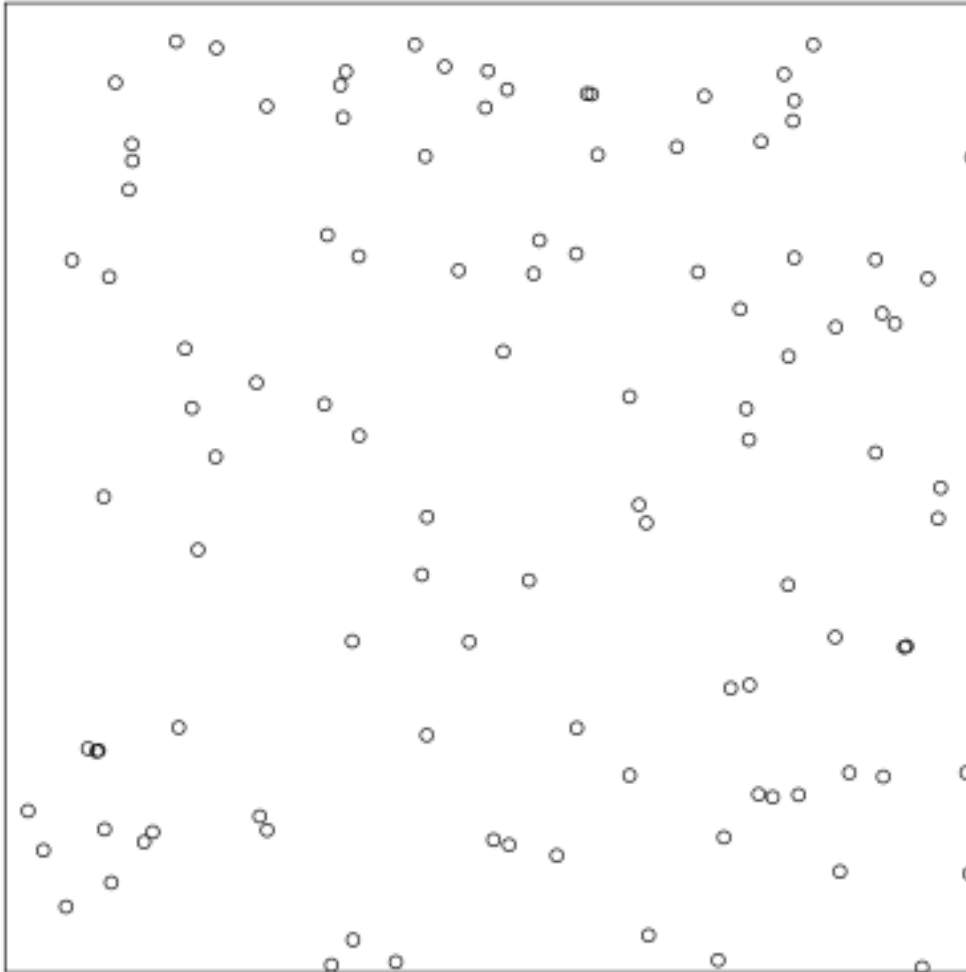
What is the role of the consumer?

What is the role of the investor?

What is the role of the employee?

What is the role of the community?

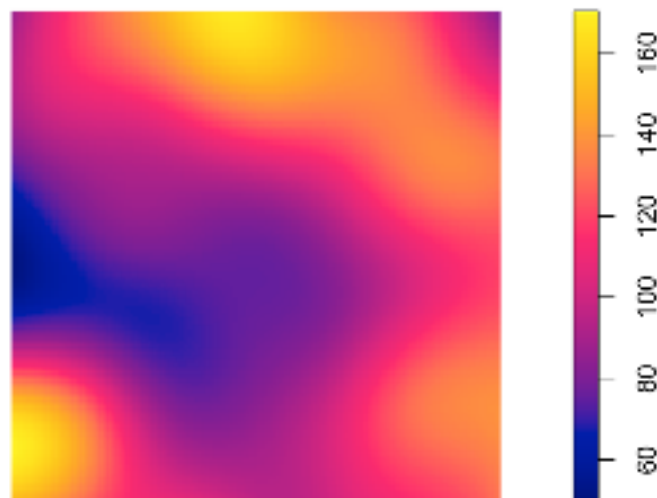
Homogeneous Poisson Process (CSR)



- Constant intensity
- Points are independent
- Density is a crude estimate of intensity

Revisting CSR

Estimated Intensity (HPP)

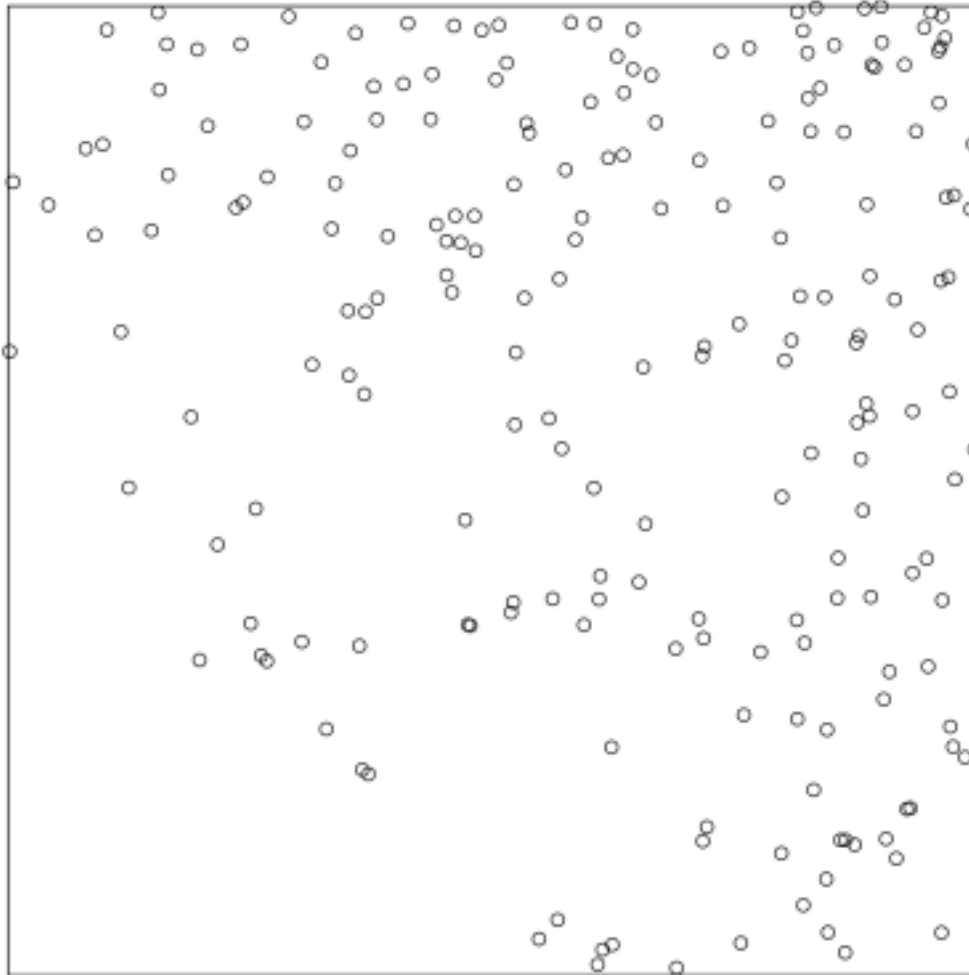


Density

0	0	100	0	0	0	0	300	200	100
200	300	100	100	100	200	0	100	300	100
100	100	0	100	100	0	0	0	200	0
200	100	200	0	200	0	100	0	0	200
300	200	100	0	0	200	100	100	0	100
100	0	300	100	0	100	0	100	300	0
200	200	0	0	100	200	0	100	0	100
100	100	100	100	200	0	0	200	300	100
200	200	200	200	100	0	200	100	200	0
0	100	100	200	0	200	200	200	100	100

First-Order Effects

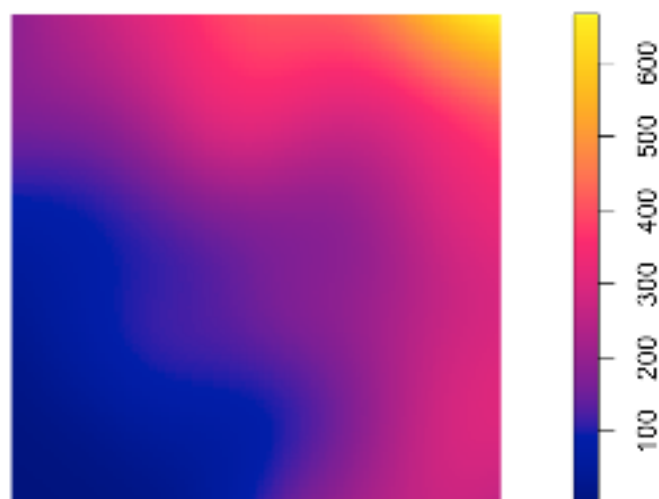
Inhomogeneous Poisson Process



- Intensity varies spatially
- Points are independent

First-Order Effects

Estimated Intensity (IPP)

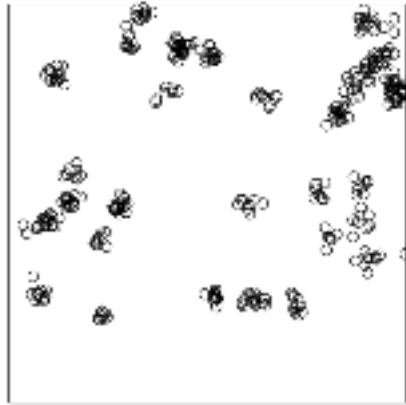


Density

0	300	200	100	0	0	0	0	0	0
500	100	100	100	200	0	100	0	0	0
200	200	200	0	0	200	300	0	0	0
300	400	200	500	100	0	200	300	0	0
500	100	800	0	0	100	200	0	0	0
400	400	300	200	300	0	400	0	0	200
700	300	100	100	100	300	200	100	0	400
200	300	200	300	0	100	300	100	300	100
1000	200	300	500	500	200	400	200	400	400
900	500	300	300	300	200	300	400	400	100

Second-Order Effects

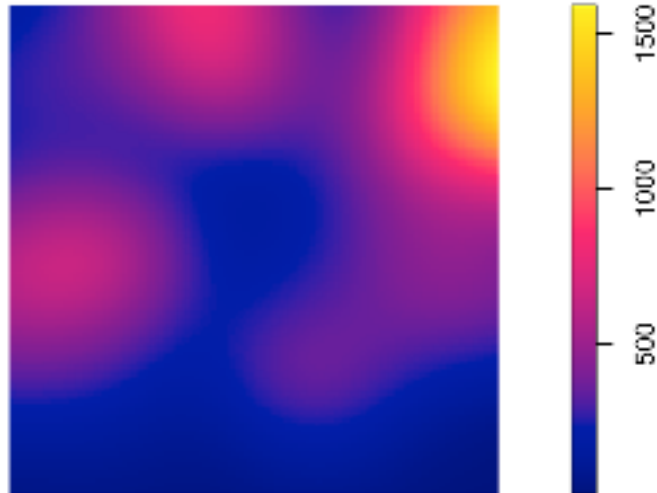
Inhomogeneous Cluster Process



- Intensity of “parents” is first-order
- Intensity of “children” is second-order

Second-Order Effects

Estimated Intensity (Clustered)



Density

0	200	0	0	0	1200	100	1000	0	0
0	1200	100	100	1700	1100	0	0	0	0
300	200	0	0	500	1900	300	800	0	0
1300	300	400	0	0	0	0	0	0	0
700	2100	300	0	0	0	0	100	0	0
0	1100	0	0	200	100	0	1500	0	0
0	0	1000	0	200	400	0	1000	0	0
0	0	0	100	900	100	100	1100	0	0
900	900	2500	0	600	1000	400	0	0	0
800	3100	2500	0	200	200	600	0	0	0

Key Points

- Poisson distribution: mean = variance = λ -> First Order
- **Homogeneous** refers to the rate, not it's outcome
- Statistical tests ask “Could this outcome be generated by a constant rate?”

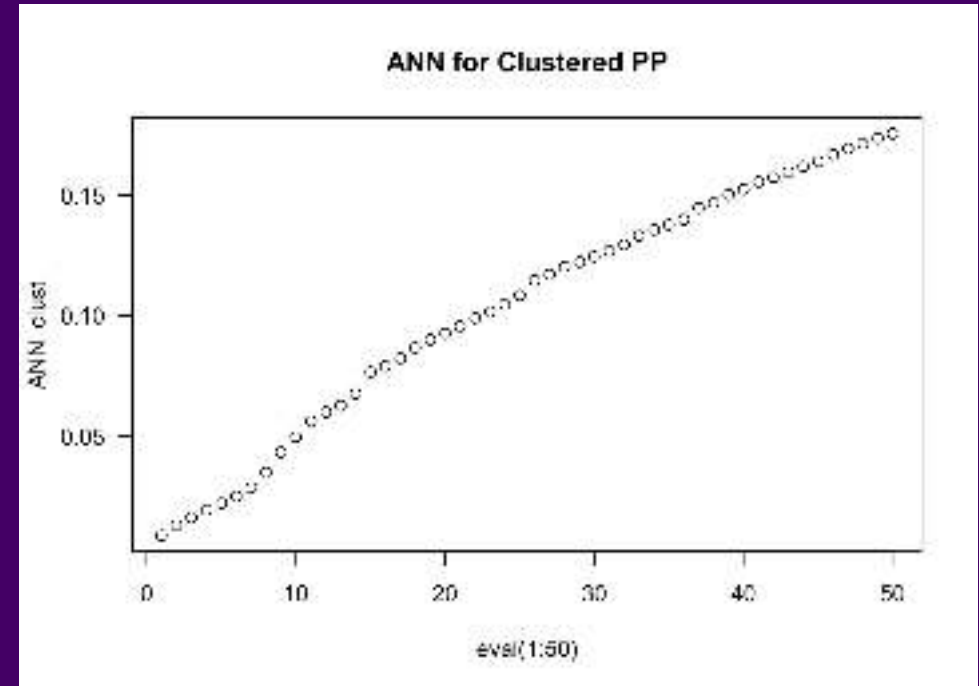
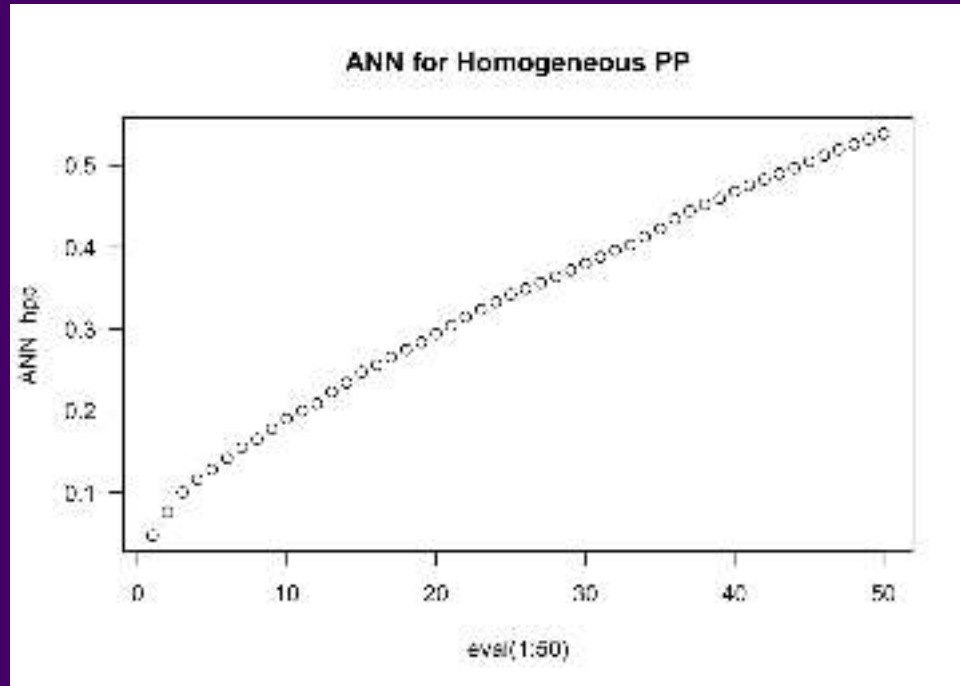
Second-Order Analysis

Distance based metrics

- Provide an estimate of the *second order* effects
- *Mean nearest-neighbor distance*:

$$\hat{d}_{\min} = \frac{\sum_{i=1}^m d_{\min}(s_i)}{n}$$

Mean Nearest Neighbor Distance



Ripley's K Function

- Nearest neighbor methods throw away a lot of information
- The K function is an alternative, based on a series of circles with increasing radius

$$K(d) = \lambda^{-1} E(N_d)$$

- We can test for clustering by comparing to the expectation:

$$K_{CSR}(d) = \pi d^2$$

- if $k(d) > K_{CSR}(d)$ then there is clustering at the scale defined by d

Ripley's K Function

- When working with a sample the distribution of K is unknown
- Estimate with

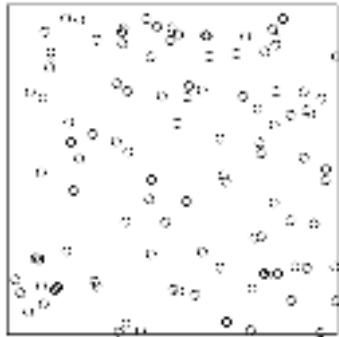
$$\hat{K}(d) = \hat{\lambda}^{-1} \sum_{i=1}^n \sum_{j=1}^n \frac{I(d_{ij} < d)}{n(n-1)}$$

where:

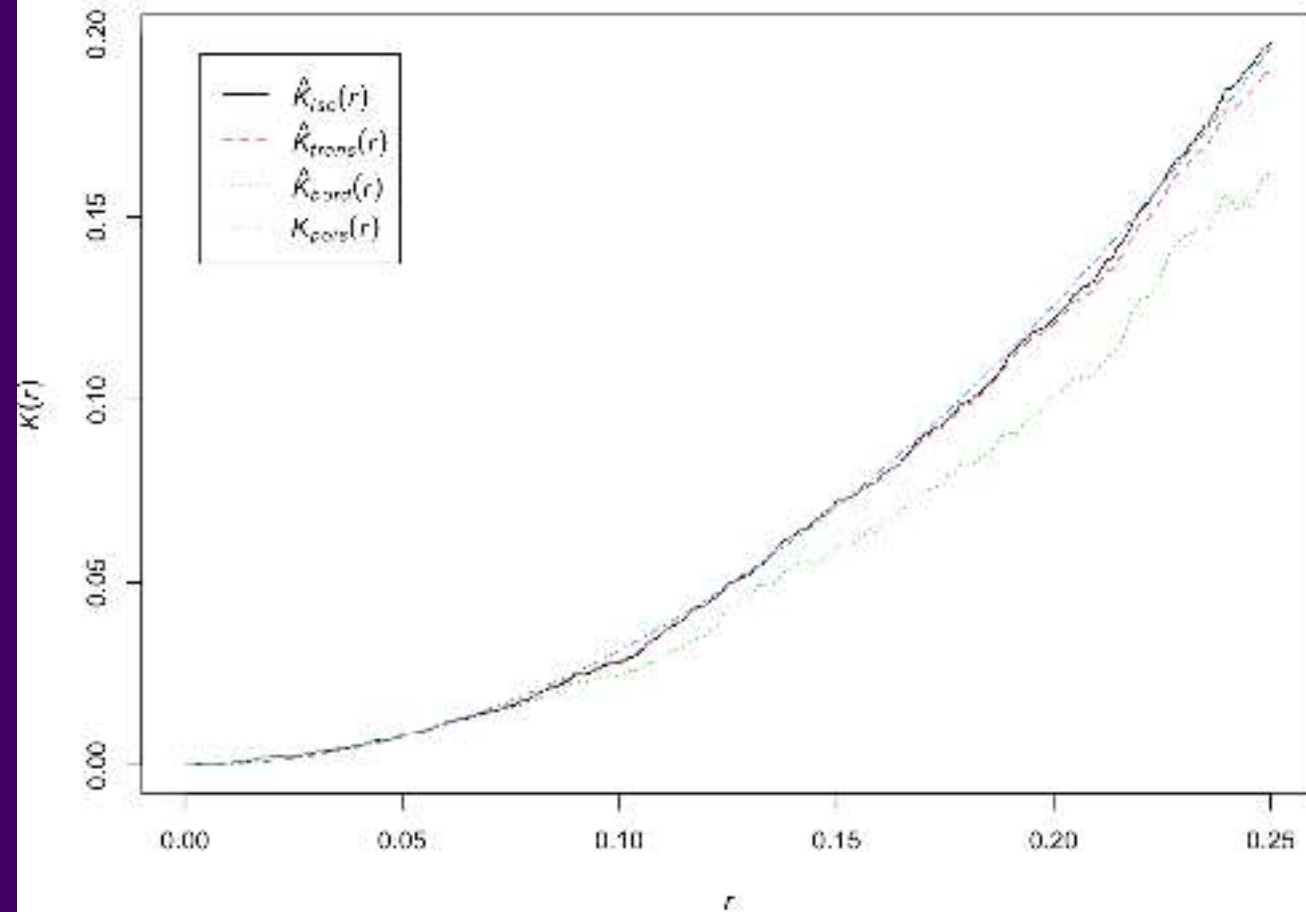
$$\hat{\lambda} = \frac{n}{|A|}$$

K for HPP

Homogeneous Poisson Process (C5R)

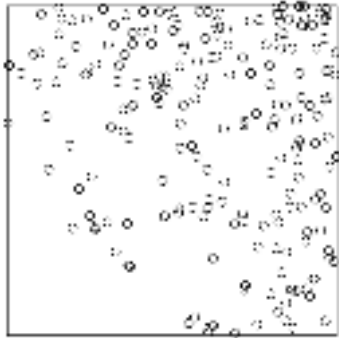


K: Homogeneous Poisson

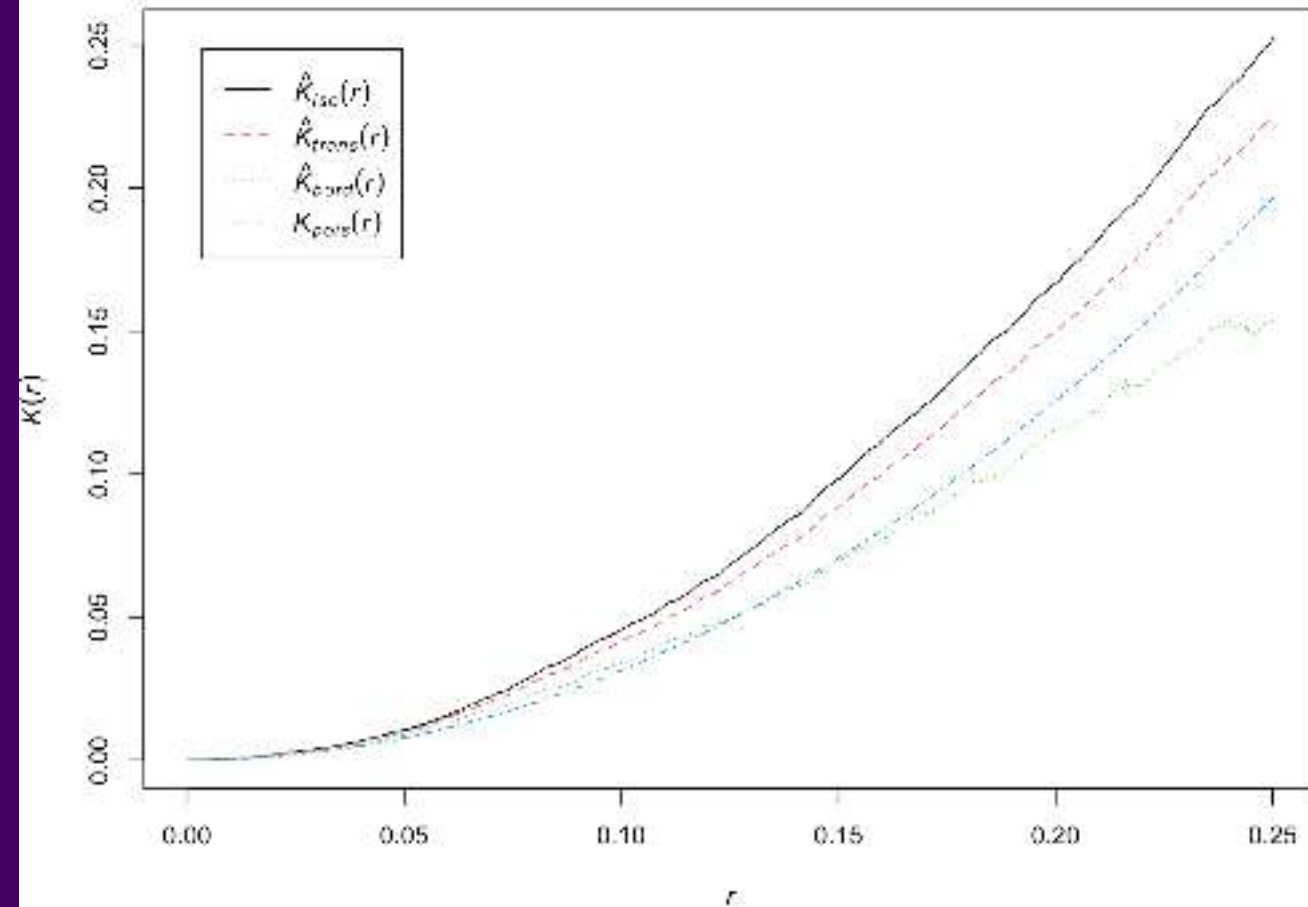


K for IPP

Inhomogeneous Poisson Process (CSR)

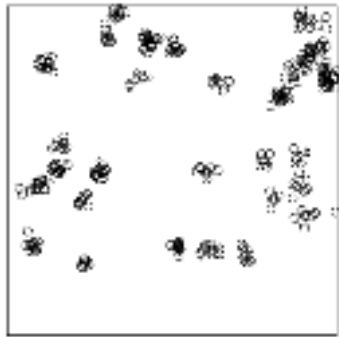


K: Inhomogeneous Poisson

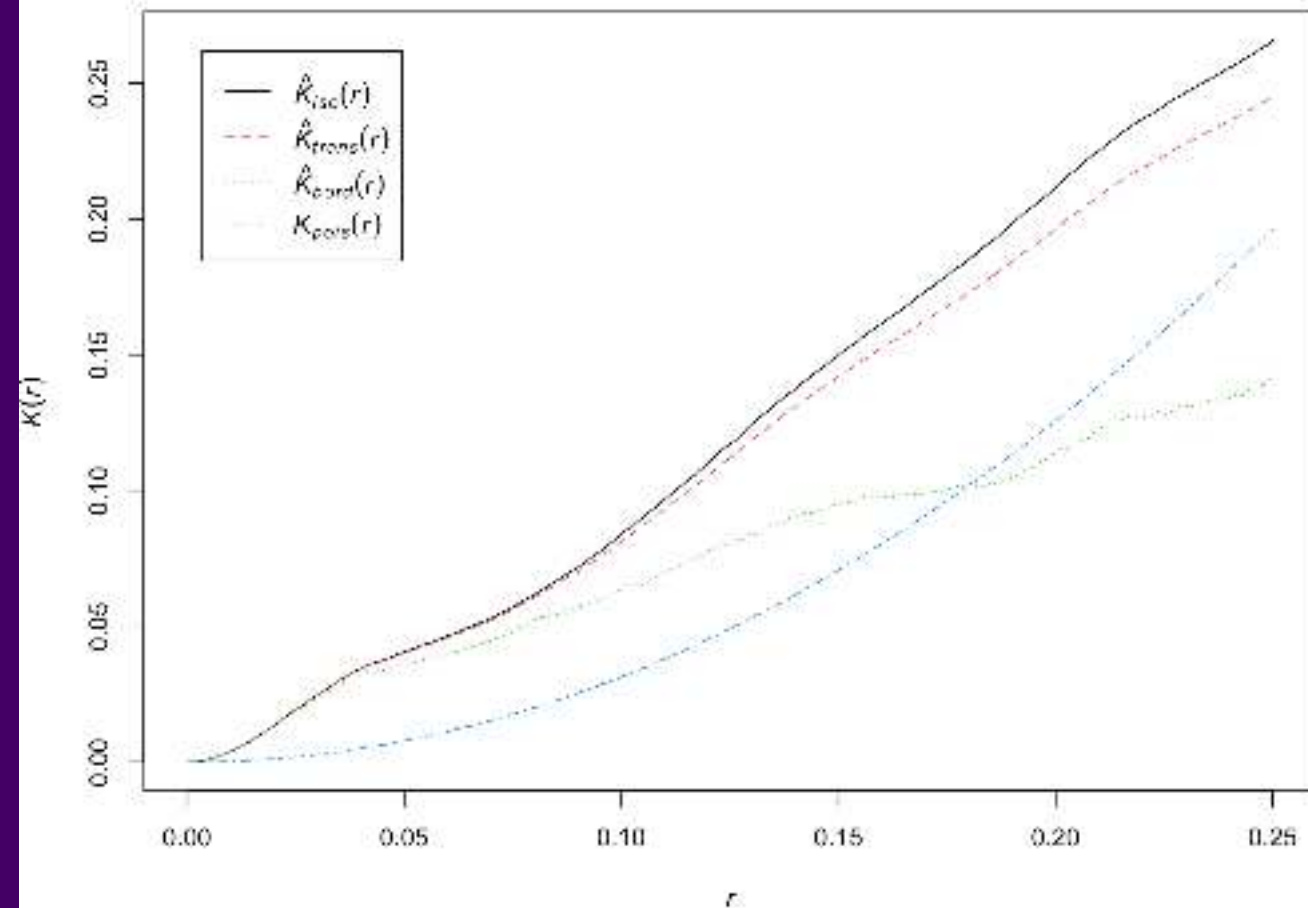


K for Clustered Processes

Clustered Poisson Process (CSR)



K: Clustered Poisson



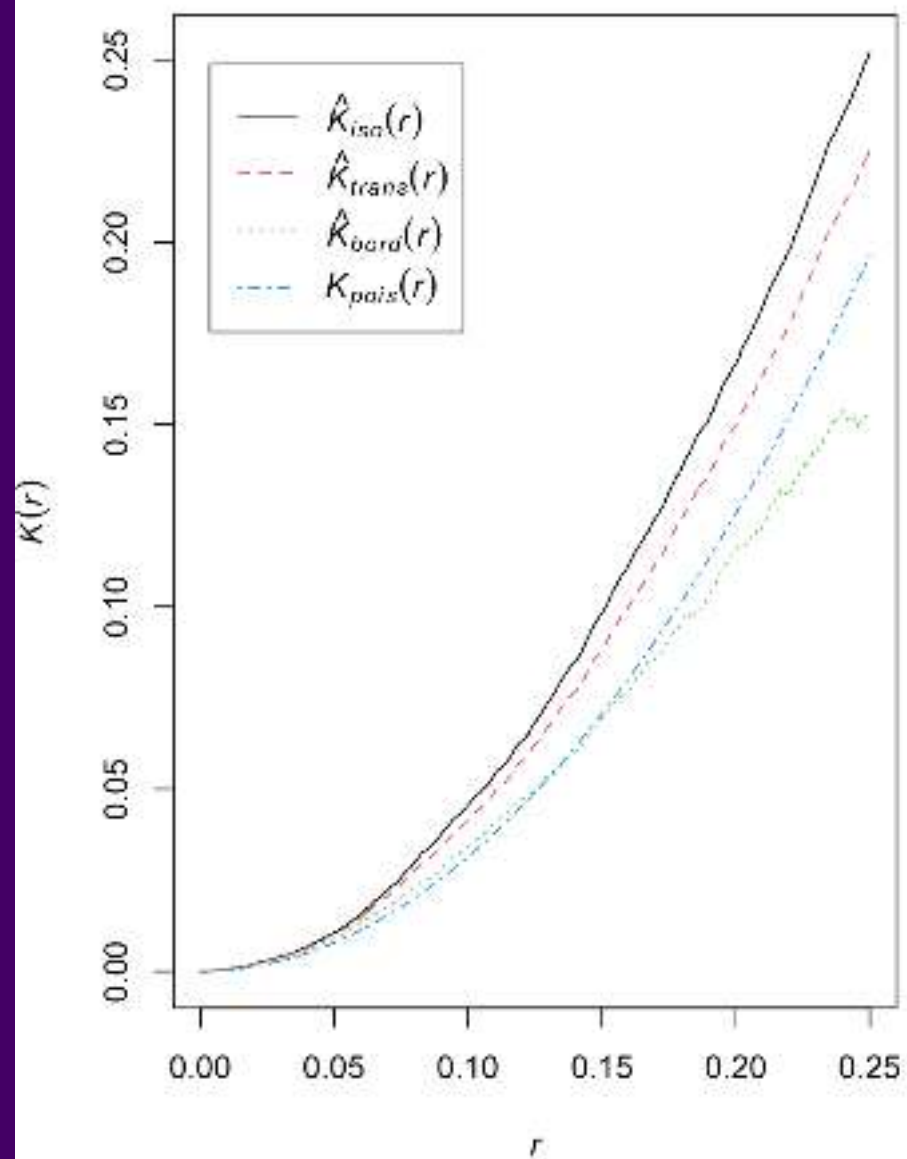
Adjustments for K

- K is sensitive to edge effects (window size)
(**correction** argument to **Kest**)
- K assumes HPP (**Kinhom** function instead of **Kest**)

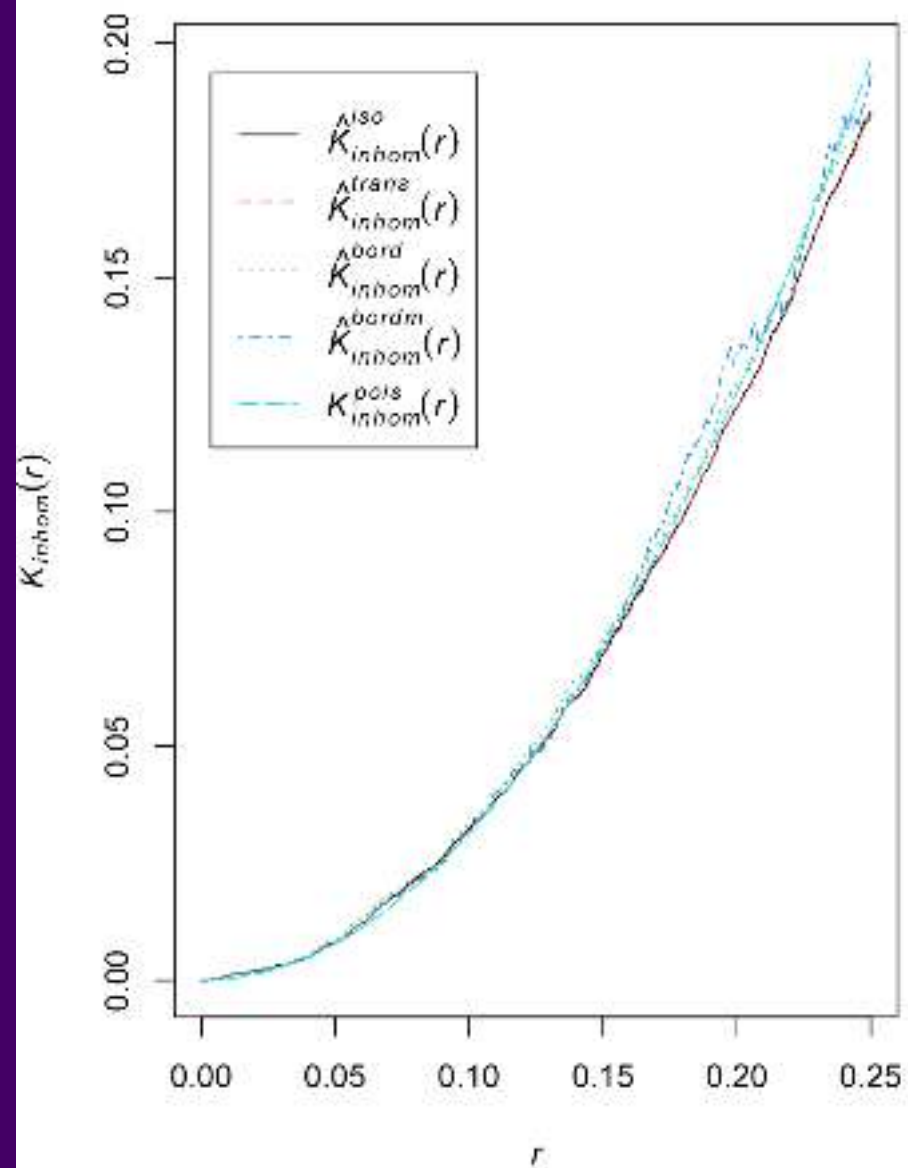
K vs K_{inhom} for IPP



K: Inhomogeneous Poisson

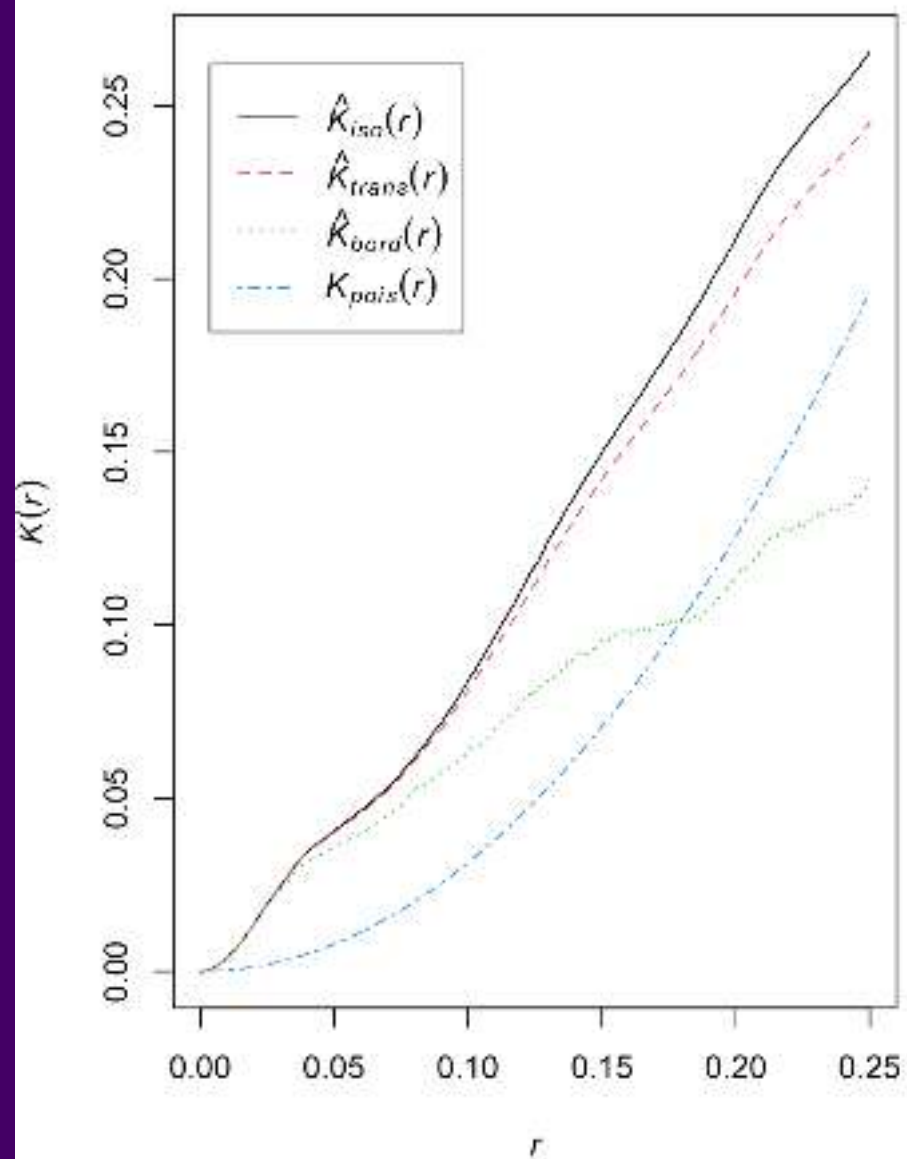


K_{inhom}: Inhomogeneous Poisson

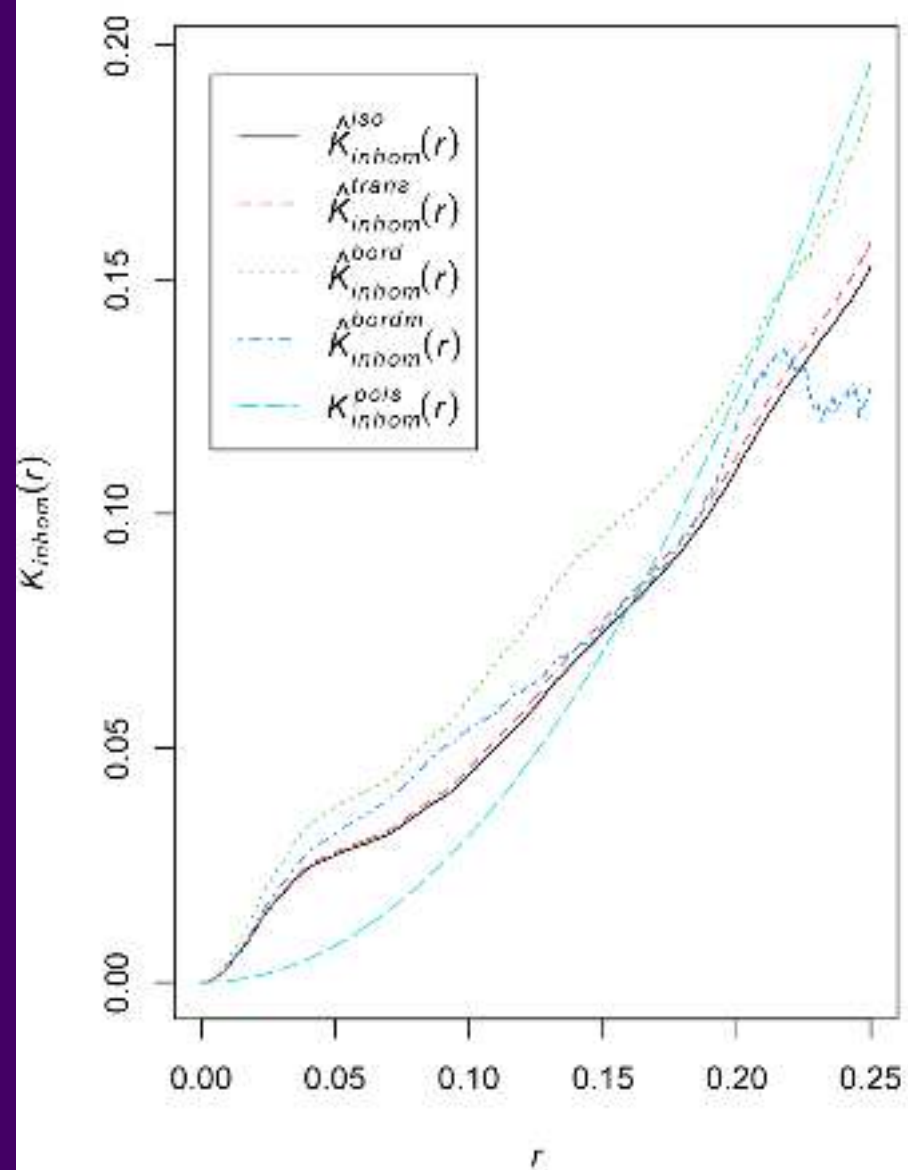


K vs K_{inhom} for Clustered PP

K: Clustered Poisson

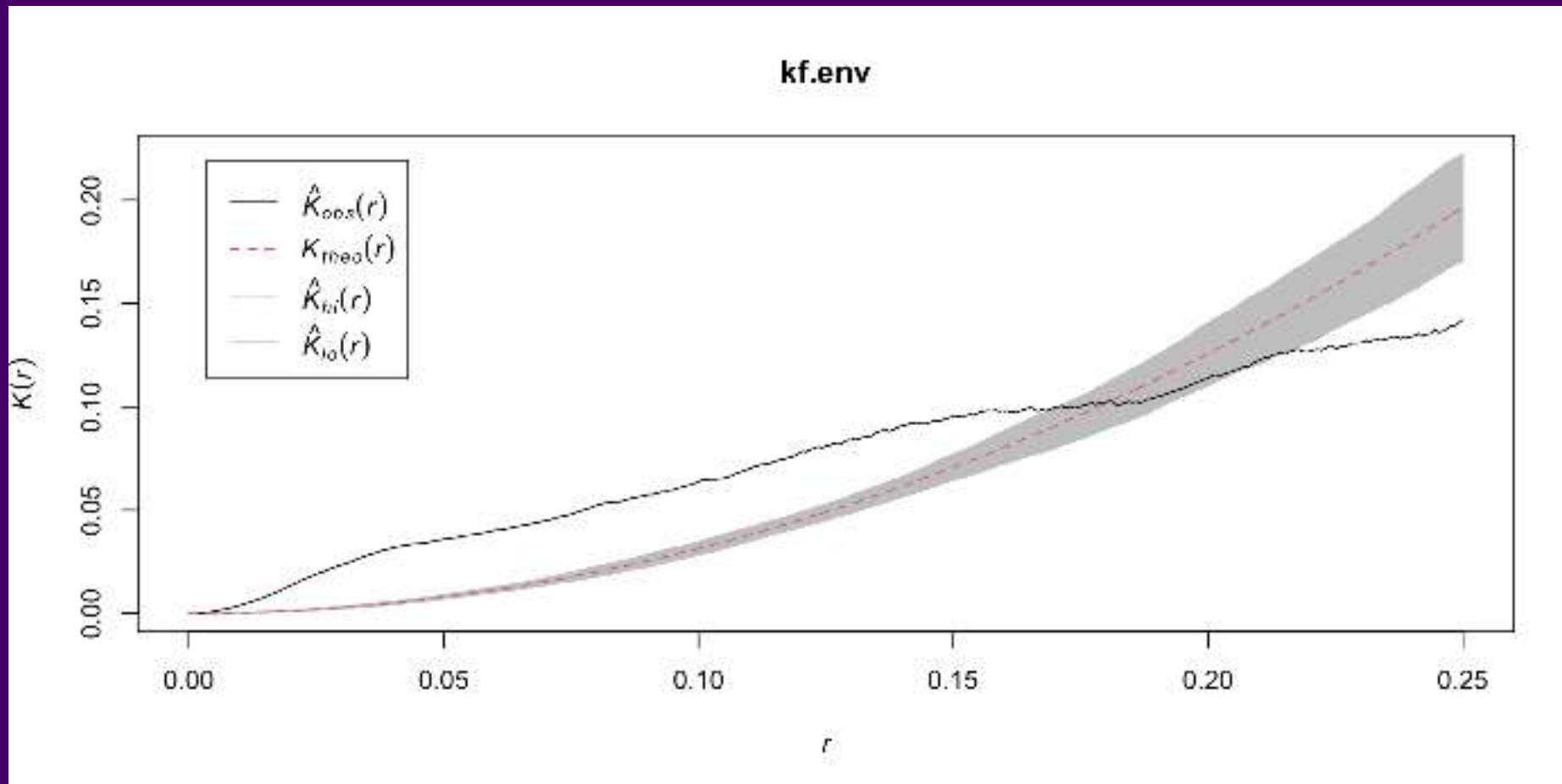


K_{inhom}: Clustered Poisson



Ripley's K Function

- accounting for variation in d



Take-Aways

- Density and Intensity for first-order effects
- Distance for second-order effects
- Linkages for kriging and interpolation

